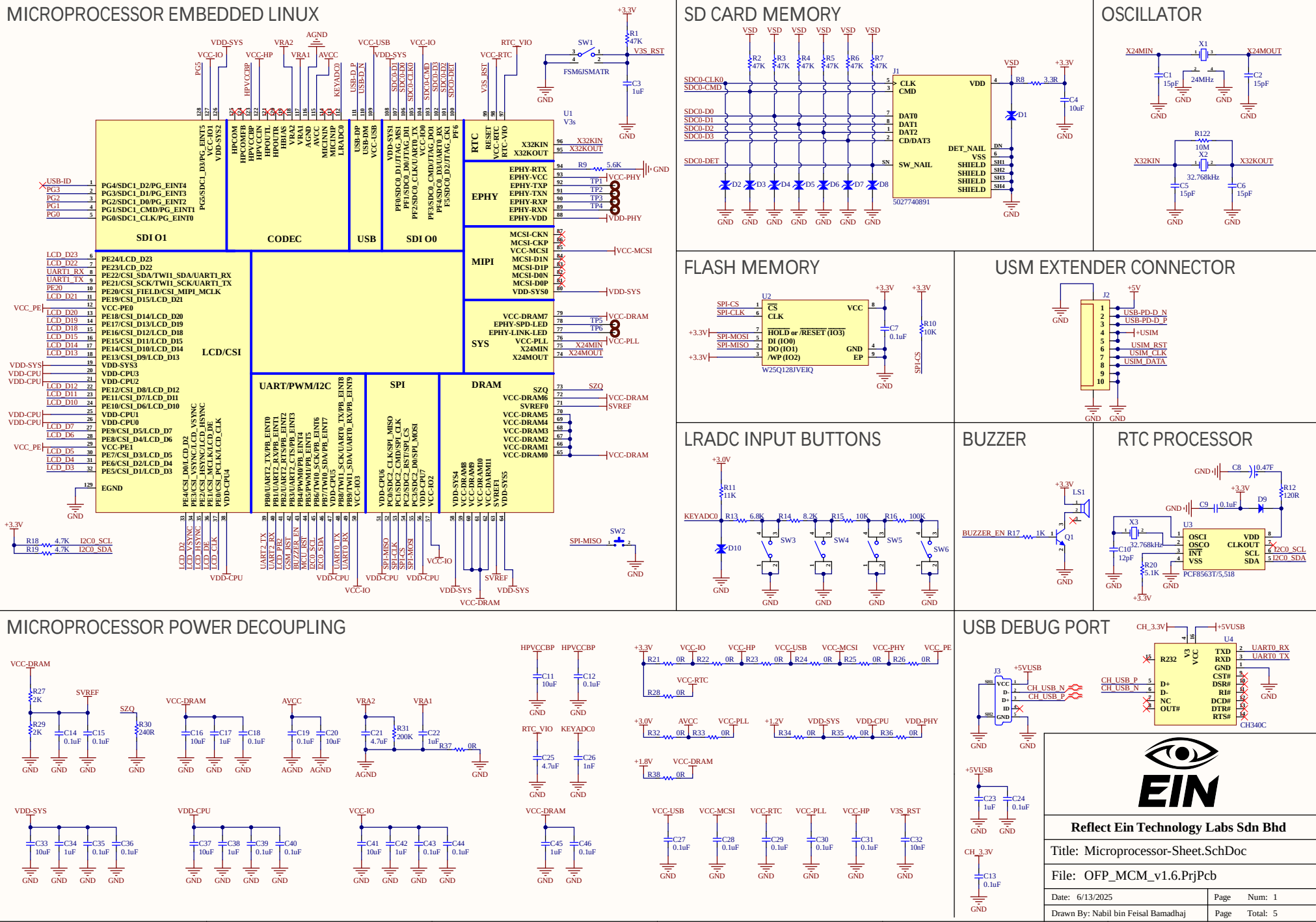
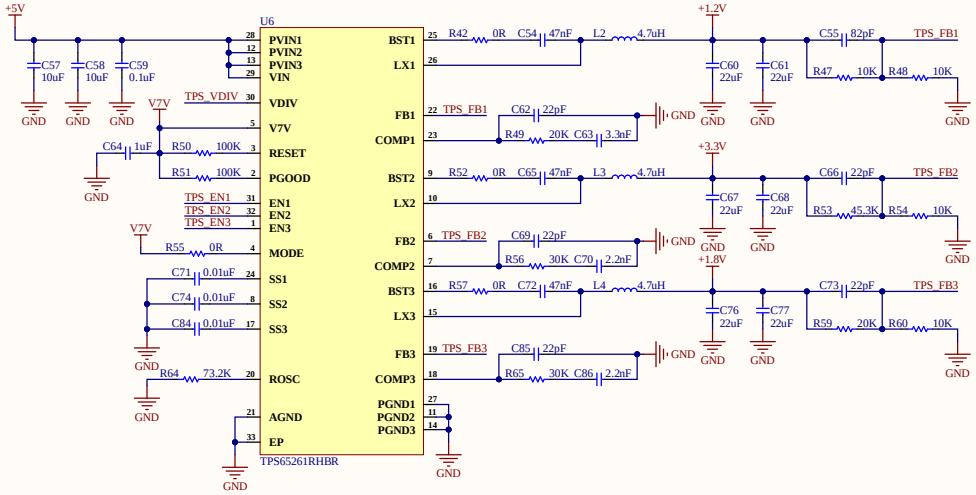

MICROPROCESSOR EMBEDDED LINUX
SD CARD MEMORY
OSCILLATOR
FLASH MEMORY
USM EXTENDER CONNECTOR
LRADC INPUT BUTTONS
BUZZER
RTC PROCESSOR
MICROPROCESSOR POWER DECOUPLING
USB DEBUG PORT
SDI O1
CODEC
USB
SDI O0
EPHY
RTC
MIPI
SYS
LCD/CSI
UART/PWM/I2C
SPI
DRAM
W25Q128JVEIQ
PCF8563T/5,518
CH340C
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Date: 6/13/2025
Drawn By: Nabil bin Feisal Bamadhaj

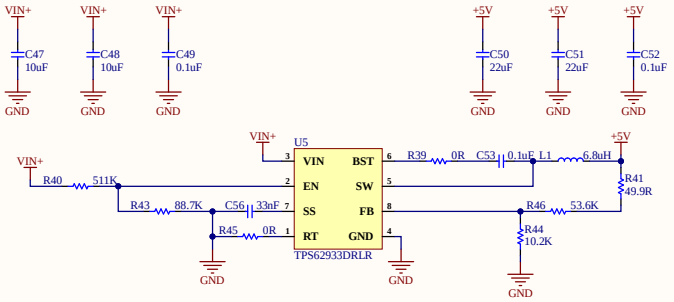
5V TO 3.3V BUCK CONVERTER

5V TO 1.8V BUCK CONVERTER

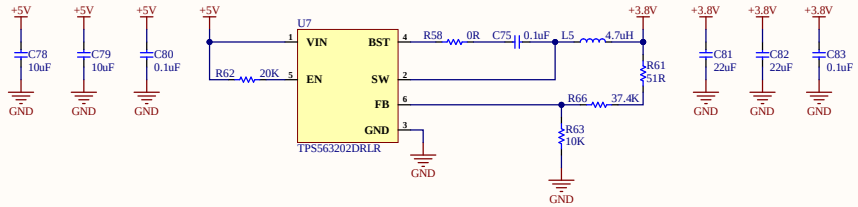
5V TO 1.2V BUCK CONVERTER



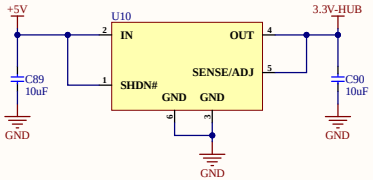
12V TO 5V BUCK CONVERTER (SYSTEM)



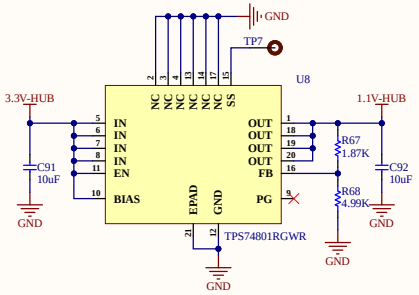
5V TO 3.8V BUCK CONVERTER (GSM)



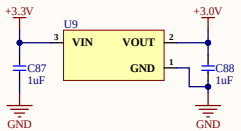
5V TO 3.3V LDO USB HUB



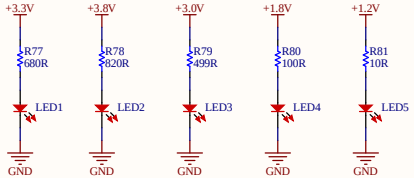
3.3V TO 1.1V LDO USB HUB

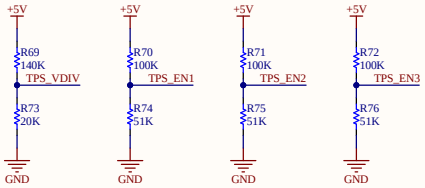


3.3V TO 3.0V LDO (MPU)



POWER INDICATORS

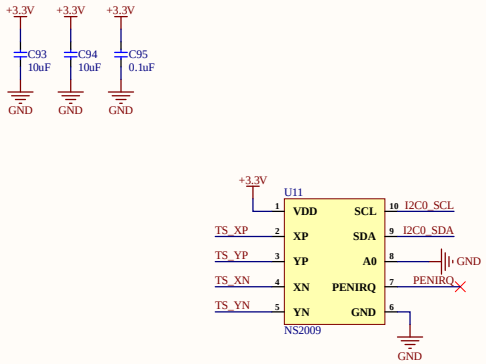




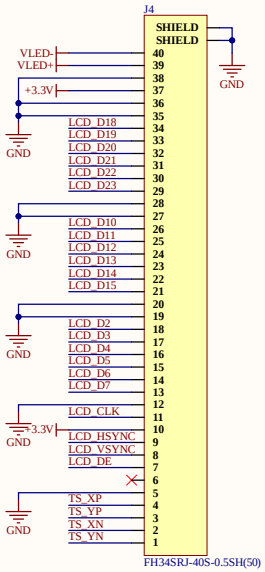
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Drawn By: Nabil bin Feisal Bamadhaj	Page Total: 5

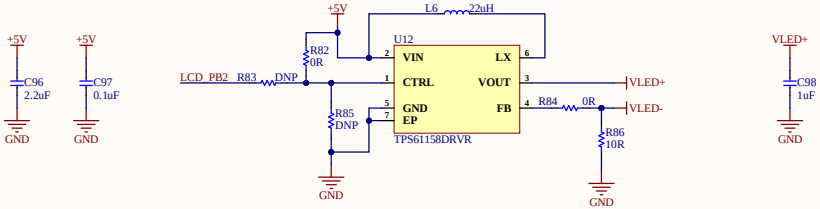
RGB LCD RESISTIVE TOUCH CONTROLLER



RGB LCD FPC/FFC CONNECTOR

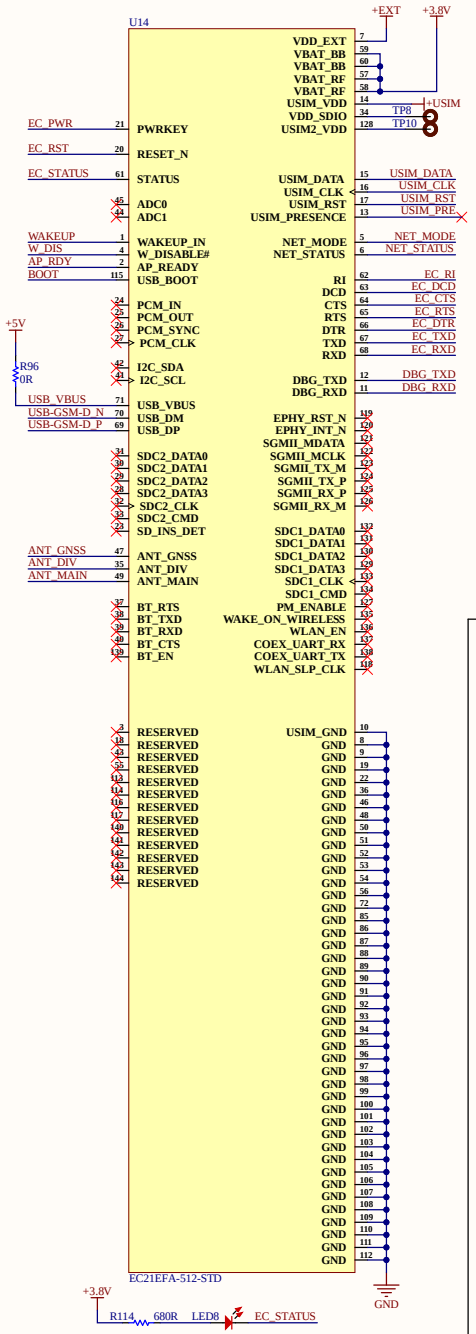


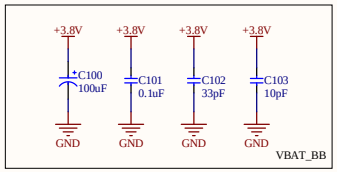
RGB LCD POWER CONTROLLER

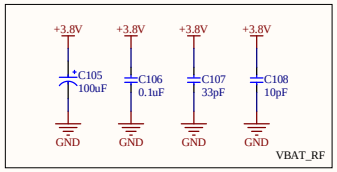


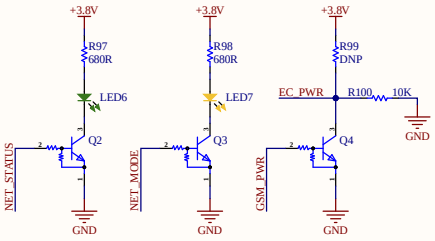
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EC21 GSM STANDARD MODULE

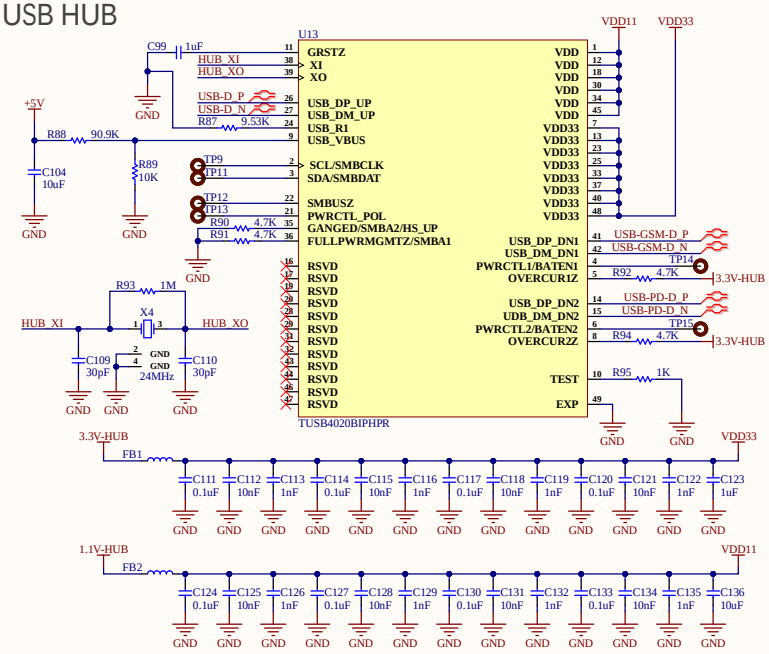




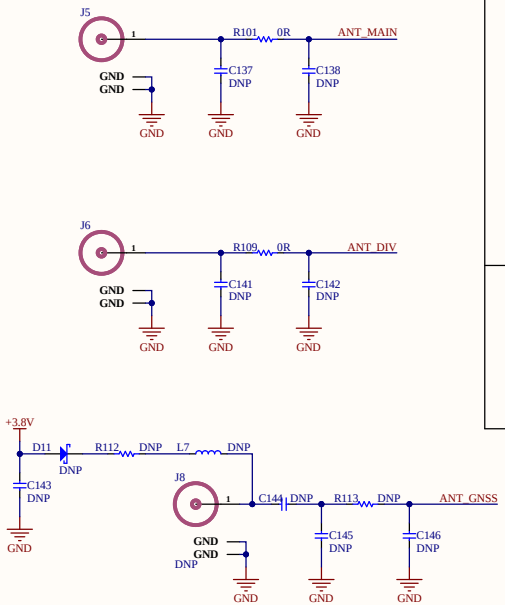




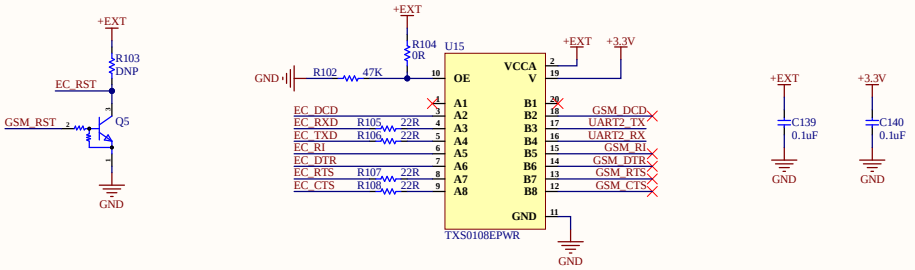
GSM USB HUB



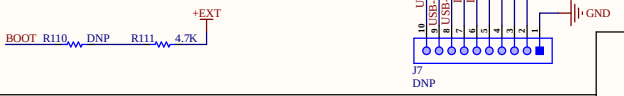
GSM ANTENNA CONNECTORS



GSM 1.8V TO 3.3V LEVEL SHIFTER



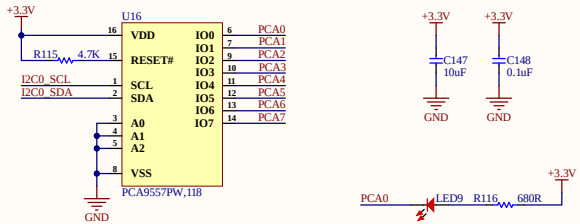
GSM OTHER UNUSED PINS



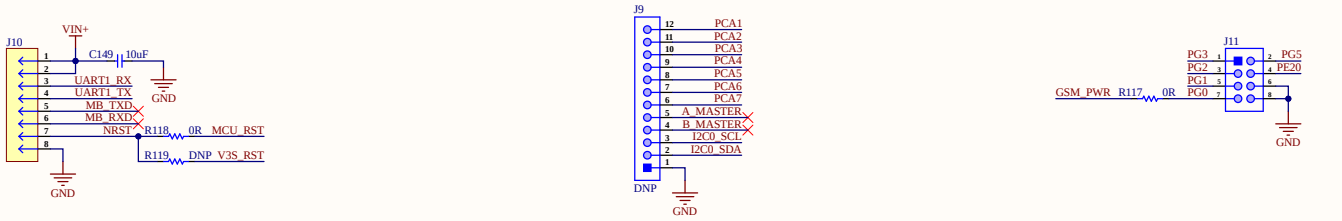
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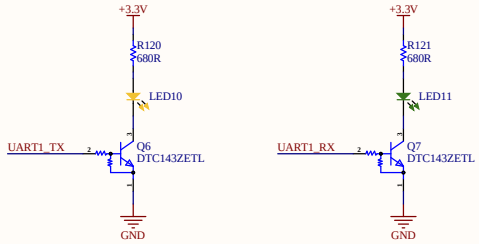
IO EXPANDER



CONNECTORS



TX / RX LED INDICATOR



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File: OFP_MCM_v1.6.PrjPcb	
Date: 6/13/2025	Page Num: 5
Drawn By: Nabil bin Feisal Bamadhaj	Page Total: 5

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.203mm	4.4	
2	Layer 1	CF-004	0.035mm		
	Dielectric 1	FR-4	1.030mm	4.6	
3	Layer 2	CF-004	0.035mm		
	Dielectric 3	PP-006	0.203mm	4.4	
4	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				

Total board thickness: 1.596mm

